

PHYSICS

SECTION-1

1) **Assertion :** In a stationary wave there is no transfer of energy.

Reason : The distance between two consecutive nodes or antinodes is $\lambda/2$.

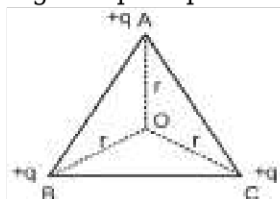
- (1) If both Assertion and Reason are true and Reason is the correct explanation of the Assertion.
- (2) If both Assertion and Reason are true but Reason is not the correct explanation of the Assertion.
- (3) If Assertion is true but Reason is false.
- (4) If both Assertion and Reason are false.

2)

V_H and V_D denote the speed of sound in humid air and dry air respectively. We find that

- (1) $V_H > V_D$
- (2) $V_H = V_D$
- (3) $V_H < V_D$
- (4) None of these

3) ABC is an equilateral triangle. Charges $+q$ are placed at each corner as shown in figure. The



electric intensity at centre O will be

- (1) $\frac{1}{4\pi\epsilon_0} \frac{q}{r}$
- (2) $\frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$
- (3) $\frac{1}{4\pi\epsilon_0} \frac{3q}{r^2}$
- (4) Zero

4) A conducting sphere of radius R is given a charge Q. The electric potential and the electric field at the centre of the sphere respectively are:

- (1) Zero and $\frac{Q}{4\pi\epsilon_0 R^2}$

(2) $\frac{Q}{4\pi\epsilon_0 R}$ and Zero

(3) $\frac{Q}{4\pi\epsilon_0 R}$ and $\frac{Q}{4\pi\epsilon_0 R^2}$

(4) Both are zero.

5) The unit of intensity of electric field is :-

(1) Newton/Coulomb

(2) Joule/Coulomb

(3) Volt-metre

(4) Newton/metre

6) Two capacitors C_1 and C_2 are charged to 120 V and 200 V, respectively. It is found that by connecting them together, the potential on each one can be made zero. Then,

(1) $5C_1 = 3C_2$

(2) $3C_1 = 5C_2$

(3) $3C_1 + 5C_2 = 0$

(4) $9C_1 = 4C_2$

7) Electrostatic force and gravitational force between two bodies differ in which respect :

(1) Conservative force

(2) Central force

(3) Principle of superposition

(4) Dependence on the intervening medium

8) The time period of revolution of a charge q_1 and of mass m moving in a circular path of radius r due to coulomb force of attraction with another charge q_2 at its centre is :

(1) $\sqrt{\frac{16\pi\epsilon_0 mr^3}{q_1 q_2}}$

(2) $\sqrt{\frac{8\pi^2\epsilon_0 mr^3}{q_1 q_2}}$

(3) $\sqrt{\frac{\epsilon_0 mr^3}{16 q_1 q_2}}$

(4) $\sqrt{\frac{16\pi^3\epsilon_0 mr^3}{q_1 q_2}}$

9) Two isolated metallic spheres of radii 2 cm and 4 cm are given equal charge, then the ratio of charge density on the surfaces of the spheres will be ?

(1) 1 : 2

- (2) 4 : 1
 (3) 8 : 1
 (4) 1 : 4

10) What will be the ratio of electric field at a point on the axis and an equidistant point on the equatorial line of a short dipole :

- (1) 1 : 2
 (2) 2 : 1
 (3) 4 : 1
 (4) 1 : 4

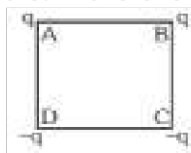
11) If specific resistance of a potentiometer wire is $10^{-7} \Omega \text{ m}$ and current flow through it is 0.1 A, cross-sectional area of wire is 10^{-6} m^2 then potential gradient will be :-

- (1) 10^{-2} volt/m
 (2) 10^{-4} volt/m
 (3) 10^{-6} volt/m
 (4) 10^{-8} volt/m

12) Newton's formula for the velocity of sound in gas is (p = pressure, ρ = density of air) :

- (1) $v = \sqrt{\frac{p}{\rho}}$
 (2) $v = \frac{2}{3} \sqrt{\frac{p}{\rho}}$
 (3) $v = \sqrt{\frac{\rho}{p}}$
 (4) $v = \sqrt{\frac{2p}{\rho}}$

13) Charges are placed on the vertices of a square as shown. Let $\vec{E}$ be the electric field and V the potential at the centre. If the charges on A and B are interchanged with those on D and C respectively, then -



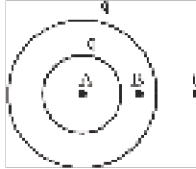
respectively, then -

- (1) $\vec{E}$ remain unchanged, V changes
 (2) both $\vec{E}$ and V change
 (3) $\vec{E}$ and V remain unchanged
 (4) $\vec{E}$ changes and V remain unchanged

14) Dimensions of permittivity of free space is :-

- (1) $[M^{-1}L^{-3}T^4A^2]$
- (2) $[M^{-1}L^3T^4A^2]$
- (3) $[M^{-1}L^{-3}T^{-4}A^2]$
- (4) $[M^{-1}L^3T^{-4}A^2]$

15) Two concentric spherical shells have charges $+q$ and $-q$ as shown in figure. Choose the wrong



options :-

- (1) At A electric field is zero, but electric potential is non-zero
- (2) At B electric field and electric potential both are non-zero
- (3) At C electric field is zero but electric potential is non-zero
- (4) At A electric field and electric potential both are zero

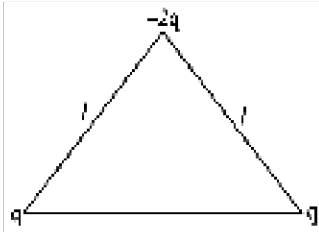
16) Match Column-I with Column -II. (σ = surface charge density, R = radius of sphere, ρ = volume charge density):

	Column-I		Column-II
I.	Electric field due to infinite plane sheet of charge	a.	0
II.	Electric field due to infinite conducting plane sheet of uniform thickness	b.	σ $2\epsilon_0$
III.	Electric field due to Non-conducting charged solid sphere at its surface	c.	$R\rho$ $3\epsilon_0$
IV.	Electric field due to Non-conducting charged solid sphere at its centre	d.	ρ ϵ_0

- (1) I-c, II-d, III-a, IV-b
- (2) I-b, II-d, III-c, IV-a
- (3) I-b, II-d, III-a, IV-c
- (4) I-b, II-a, III-d, IV-c

17)

The dipole moment of the given system is (triangle is equilateral):



- (1) $\sqrt{3}ql$ along perpendicular bisector of $q - q$ line.
- (2) $2ql$ along perpendicular bisector of $q - q$ line.
- (3) $ql\sqrt{2}$ along perpendicular bisector of $q - q$ line.
- (4) 0

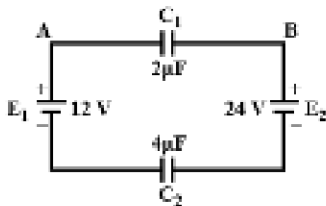
18) Minimum number of $2\mu\text{F}$ and 250 V capacitors used to make a combination of $12\mu\text{F}$ and 500V are :-

- (1) 48
- (2) 24
- (3) 12
- (4) 6

19) The tension in a piano wire is 10 N . The tension in a piano wire to produce a node of double frequency is :

- (1) 20 N
- (2) 40 N
- (3) 10 N
- (4) 120 N

20) Two capacitors C_1 and C_2 are connected in a circuit as shown in figure. The potential difference



$(V_A - V_B)$ is :-

- (1) 8 V
- (2) -8 V
- (3) 12 V
- (4) -12 V

21) A parallel plate capacitor has plates of area A and separation d and it is charged to a potential difference V . The charging battery is then disconnected and the plates are pulled apart until their separation is $2d$. What is the work required to separate the plates ?

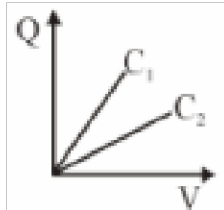
- (1) $2\epsilon_0 AV^2/d$
- (2) $\epsilon_0 AV^2/d$
- (3) $3\epsilon_0 AV^2/2d$

(4) $\epsilon_0 AV^2/2d$

22) Three capacitors of capacitances $3\ \mu\text{F}$, $9\ \mu\text{F}$ and $18\ \mu\text{F}$ are connected once in series and another time in parallel. The ratio of equivalent capacitance in the two cases (C_s/C_p) will be

- (1) 1 : 15
- (2) 15 : 1
- (3) 1 : 1
- (4) 1 : 3

23) The following graph shows the variation of charge Q and voltage V for two capacitors C_1 and C_2 .



Choose the correct relation.

- (1) $C_1 > C_2$
- (2) $C_1 < C_2$
- (3) $C_1 = C_2$
- (4) cannot be determined

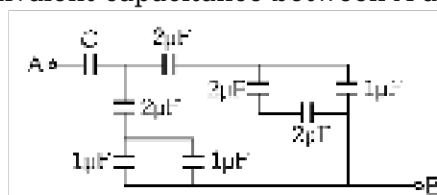
24) A transverse waves is represented by $y = A \sin (\omega t - kx)$. For what value of the wavelength is the wave velocity equal to the maximum particle velocity ?

- (1) A
- (2) $\frac{\pi A}{2}$
- (3) πA
- (4) $2\pi A$

25) A capacitor of capacitance $500\ \mu\text{F}$ is charged at the rate of $100\ \mu\text{C/s}$. The time in which the potential difference will become 20V is :-

- (1) 100 s
- (2) 50 s
- (3) 20 s
- (4) 10 s

26) In the given diagram the equivalent capacitance between A and B is $1.0\ \mu\text{F}$. What is the



capacitance of the condenser C.

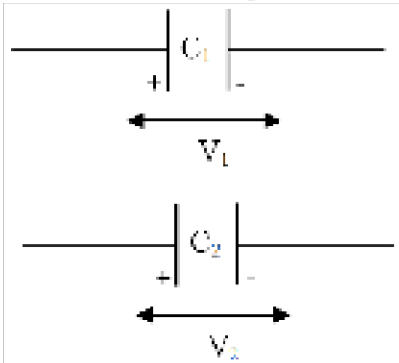
- (1) $2\ \mu\text{F}$

- (2) $5 \mu\text{F}$
 (3) $6 \mu\text{F}$
 (4) $1 \mu\text{F}$

27) Two metal spheres, one of radius R and the other of radius $2R$ respectively have the same surface charge density σ . They are brought in contact and separated. What will be the new surface charge densities on them?

- (1) $\sigma_1 = \frac{5}{6}\sigma, \sigma_2 = \frac{5}{2}\sigma$
 (2) $\sigma_1 = \frac{5}{2}\sigma, \sigma_2 = \frac{5}{6}\sigma$
 (3) $\sigma_1 = \frac{5}{2}\sigma, \sigma_2 = \frac{5}{3}\sigma$
 (4) $\sigma_1 = \frac{5}{3}\sigma, \sigma_2 = \frac{5}{6}\sigma$

28) Find common potential if plates of same polarities are connected with each other



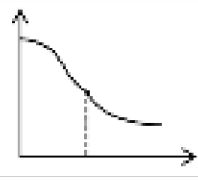
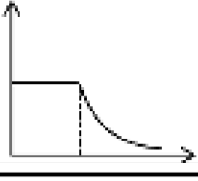
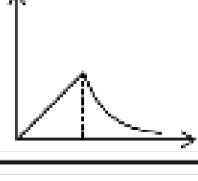
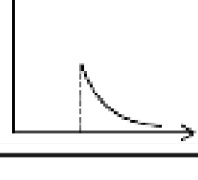
- (1) $\frac{C_1 + C_2}{C_1 V_1 + C_2 V_2}$
 (2) $\frac{C_1 C_2}{C_1 V_1 - C_2 V_2}$
 (3) $\frac{C_1 C_2}{C_1 V_1 + C_2 V_2}$
 (4) $\frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$

29) A charged particle of mass m and charge q is released from rest in an electric field of constant magnitude E . The KE of the particle after time t is

- (1) $\frac{Eq^2 m}{2t^2}$
 (2) $\frac{2E^2 t^2}{mq}$
 (3) $\frac{E^2 q^2 t^2}{2m}$
 (4) $\frac{Eqm}{2t}$

30) Match the columns

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	Column-A		Column-B
A.	Variation of electric field with distance from centre in case of uniformly charged spherical shell	I.	
B.	Variation of electric potential in (A)	II.	
C.	Variation of electric field with distance from centre in case of uniformly charged non-conducting sphere	III.	
D.	Variation of electric potential in (C)	IV.	

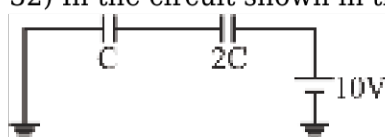
- | | A | B | C | D |
|-----|----|-----|-----|-----|
| (1) | I | II | III | IV |
| (2) | IV | III | I | II |
| (3) | IV | II | III | I |
| (4) | II | I | IV | III |

- (1) 1
(2) 2
(3) 3
(4) 4

31) A standing wave is represented by $y = a \sin(100t) \cos(0.01x)$ where t is in seconds and x is in metres. Then the velocity of the wave is :

- (1) 10^{-2} m/s
(2) 10^{-4} m/s
(3) 1 m/s
(4) 10^4 m/s

32) In the circuit shown in the figure, $C = 6\mu\text{F}$. The charge stored in the capacitor of capacity C is :-



- (1) zero
(2) $90\mu\text{C}$

(3) $40\mu\text{C}$ (4) $60\mu\text{C}$

33) Match the pairs:

	Capacitor		Capacitance
(A)	Cylindrical capacitor	(i)	$4\pi\epsilon_0 R$
(B)	Spherical capacitor	(ii)	$\frac{KA\epsilon_0}{d}$
(C)	Parallel plate capacitor having dielectric between its plates	(iii)	$\frac{2\pi\epsilon_0\ell}{\ell n \frac{r_2}{r_1}}$
(D)	Isolated spherical Conductor	(iv)	$\frac{4\pi\epsilon_0 r_1 r_2}{r_2 - r_1}$

(1) A-(iii) B-(iv) C-(ii) D-(i)

(2) A-(i) B-(ii) C-(iii) D-(iv)

(3) A-(iii) B-(i) C-(iv) D-(ii)

(4) A-(iv) B-(iii) C-(i) D-(ii)

34) Two identical capacitors, have the same capacitance C . One of them is charged to potential V_1 and the other to V_2 . The negative ends of the capacitors are connected together. When the positive ends are also connected, the decrease in energy of the combined system is :-

(1) $\frac{1}{4}C (V_1^2 - V_2^2)$

(2) $\frac{1}{4}C (V_1^2 + V_2^2)$

(3) $\frac{1}{4}C (V_1 - V_2)^2$

(4) $\frac{1}{4}C (V_1 + V_2)^2$

35) If n drops, each of capacitance C and each having same charge q , coalesce to form a single drop. then the ratio of the energy stored in the big drop to that in each small drops will be :

(1) $n : 1$

(2) $n^{\frac{1}{3}} : 1$

(3) $n^{\frac{5}{3}} : 1$

(4) $n^2 : 1$

SECTION-2

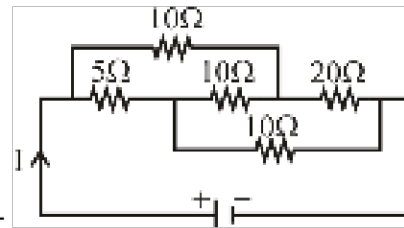
1) The ratio of speed of sound in helium and hydrogen gases at the same temperature is :

(1) $\sqrt{42} : \sqrt{25}$

(2) $\sqrt{25} : \sqrt{42}$

(3) 42 : 25

(4) 25 : 42



2) The current I drawn from the 5 volt source will be :-

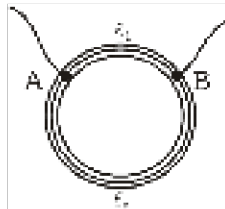
(1) 0.17 A

(2) 0.33 A

(3) 0.5 A

(4) 0.67 A

3) A ring is made of a wire having a resistance $R_0 = 12 \Omega$. Find the points A and B as shown in the figure at which a current carrying conductor should be connected so that the resistance R of the sub



circuit between these points is equal to $\frac{8}{3} \Omega$:-

(1) $\frac{\ell_1}{\ell_2} = \frac{3}{8}$

(2) $\frac{\ell_1}{\ell_2} = \frac{1}{2}$

(3) $\frac{\ell_1}{\ell_2} = \frac{5}{8}$

(4) $\frac{\ell_1}{\ell_2} = \frac{1}{3}$

4) The kirchhoff's current law ($\sum i = 0$) and voltage law ($\sum iR = \sum E$), Where the symbols have their usual meanings, are respectively based on:

(1) conservation of charge, conservation of momentum

(2) conservation of energy, conservation of charge

(3) conservation of momentum, conservation of charge

(4) conservation of charge, conservation of energy

5) A heater coil takes 4 min to boil some amount of water. Another heater will takes 6 min to boil the same amount of water with the same supply voltage. Now the two coils are connected in parallel combination then how much time will it take to boil the same amount of water. (Supply voltage is same in all cases) :-

(1) 1.2 min

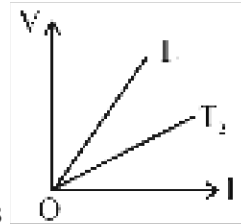
(2) 2.4 min

- (3) 3.6 min
(4) 4.8 min

6) In resonance tube two position of resonance are obtained at 15 cm and 48 cm. If the frequency of the fork is 500 Hz, the velocity of sound is:

- (1) 330 m/s
(2) 300 m/s
(3) 1000 m/s
(4) 360 m/s

7) The voltage V and current I graphs for a conductor at two different temperature T_1 and T_2 are



shown in the figure. The relation between T_1 and T_2 is

- (1) $T_1 > T_2$
(2) $T_1 < T_2$
(3) $T_1 = T_2$
(4) $T_1 = \frac{1}{T_2}$

8)

A uniform conductor of resistance R is cut into 20 equal pieces. Half of them are joined in series and the remaining half of them are connected in parallel. If the two combinations are joined in series, the effective resistance of all the pieces

is :-

- (1) R
(2) $R/2$
(3) $101R/200$
(4) $201R/200$

9) Match the Pairs

(A)	Drift velocity (v_d)	(i)	$\frac{e\tau}{m}E$
(B)	Current density (j)	(ii)	$n e v_d$
(C)	Conductivity (σ)	(iii)	$\frac{ne^2\tau}{m}$
(D)	Resistivity (ρ)	(iv)	$\frac{m}{ne^2\tau}$

- (1) A-i, B-ii, C-iii, D-iv

- (2) A-i, B-ii, C-iv, D-iii
 (3) A-i, B-iii, C-ii, D-iv
 (4) A-iv, B-iii, C-ii, D-i

10) The equation $y = 4 + 2 \sin (6t - 3x)$ represents a wave motion. Then wave speed and amplitude, respectively are

- (1) wave speed 1 unit, amplitude 6 unit
 (2) wave speed 2 unit, amplitude 2 unit
 (3) wave speed 4 unit, amplitude $\frac{1}{2}$ unit
 (4) wave speed $\frac{1}{2}$ unit, amplitude 5 unit

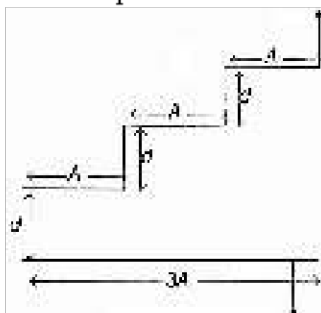
11) Which of the following statements is false ?

- (1) A rheostat can be used as a potential divider
 (2) Kirchhoff's second law represents energy conservation
 (3) Wheatstone bridge is the most sensitive when all the four resistances are of the same order of magnitude.
 (4) In a balanced wheatstone bridge if the cell and the galvanometer are exchanged, the null point is disturbed.

12) The electric field in a certain region is given by $\vec{E} = (5\hat{i})$ kV/m. The potential difference ($V_B - V_A$) between two points A and B having co-ordinates (4, 0, 0) and (10, 3, 0) respectively, is equal to (co-ordinates x, y, z are in metre)

- (1) 12 kV
 (2) 18 kV
 (3) -42 kV
 (4) -30 kV

13) The expression for the equivalent capacitance of the system shown is (A is the cross-sectional



area of one of the upper plates)

- (1) $\frac{\epsilon_0 A}{3d}$
 (2) $\frac{3A\epsilon_0}{d}$
 (3) $\frac{11A\epsilon_0}{6d}$
 (4) $\frac{6A\epsilon_0}{d}$

14) When 2A current is passed through 2Ω resistance, the potential difference across the terminals of a cell is $E/2$. The internal resistance of the cell is -

- (1) 1
- (2) 2
- (3) 3
- (4) 4

15) Two tuning forks P and Q when set vibrating, given 4 beats/s. If a prong of the fork P is filed, the beats are reduced to 2 beats/s. What is frequency of P, if that of Q is 250 Hz ?

- (1) 246 Hz
- (2) 250 Hz
- (3) 254 Hz
- (4) 252 Hz

CHEMISTRY

SECTION-1

1) The equilibrium constant of a reaction is 20.0. At equilibrium, the rate constant of forward reaction is 10. The rate constant for backward reaction is :

- (1) 0.5
- (2) 2.0
- (3) 10.0
- (4) 200.0

2) The equilibrium constant of a reaction is 300. If the volume of reaction flask is doubled, the equilibrium constant is :-

- (1) Double
- (2) Halved
- (3) Remain same
- (4) Can't predict

3) In dissociation of $\text{CaCO}_3(\text{s})$ which gives $\text{CaO}(\text{s})$ and $\text{CO}_2(\text{g})$ is favoured by -

- (1) High pressure
- (2) High conc. of CO_2
- (3) High temperature
- (4) Low temperature

4) **Assertion (A)** : A catalyst does not influence the value of equilibrium constant.

Reason (R) : Catalyst influences the rate of both forward and backward reactions equally.

- (1) Both **Assertion** and **Reason** are correct and **Reason** is correct explanation of **Assertion**.
(2) Both **Assertion** and **Reason** are correct and **Reason** is not correct explanation of **Assertion**.
(3) **Assertion** is correct but **Reason** is incorrect
(4) **Assertion** is incorrect but **Reason** is correct
- 5) Increase in temperature in a reversible equilibrium reaction favours –
- (1) Forward reaction only
(2) Backward reaction only
(3) Both forward or backward reaction
(4) Neither forward nor backward reaction
- 6) $\text{PCl}_{5(g)} \rightleftharpoons \text{PCl}_{3(g)} + \text{Cl}_{2(g)}$
In above reaction, at equilibrium condition mole fraction of PCl_5 is 0.4 and mole fraction of Cl_2 is 0.3. Then find out mole fraction of PCl_3
- (1) 0.3
(2) 0.7
(3) 0.4
(4) 0.6
- 7) 2 mole of PCl_5 were heated in a closed vessel of 2 litre capacity. At equilibrium 40% of PCl_5 dissociated into PCl_3 and Cl_2 , the value of the equilibrium constant is :-
- (1) 0.267
(2) 0.53
(3) 2.63
(4) 5.3
- 8) The $[\text{OH}^-]$ of an aqueous solution is 1×10^{-5} . The pH of the solution is :
- (1) 5
(2) 9
(3) 4.5
(4) 11
- 9) For which of the following ostwald's dilution law is applicable :-
- (1) HCl
(2) NaOH
(3) CH_3COOH
(4) H_2SO_4
- 10) The pH of 0.5 M aqueous solution of HF
($K_a = 2 \times 10^{-4}$) is :-

- (1) 2
- (2) 4
- (3) 6
- (4) 10

11) Which among the following aqueous salt solution is acidic in nature?

- (1) Sodium acetate
- (2) Potassium chloride
- (3) Ammonium sulphate
- (4) Sodium cyanide

12) Which of the following salt undergoes anionic hydrolysis ?

- (1) AlCl_3
- (2) CuSO_4
- (3) Na_2CO_3
- (4) NH_4Cl

13) Calculate pH of a salt of weak monobasic acid and weak monoacidic base having concentration 0.1M at 25°C (Given :- $\text{pK}_a = 4.8$, $\text{pK}_b = 5.2$)

- (1) 7.2
- (2) 6.8
- (3) 9.6
- (4) 7

14) $\text{M}(\text{OH})_x$ has $K_{sp} = 4 \times 10^{-12}$ and solubility 10^{-4} M. Hence, x is :-

- (1) 1
- (2) 2
- (3) 3
- (4) 4

15) $[\text{H}^+]$ in the aqueous solution of the mixture of 0.1M NaCN and 0.1M HCN ($K_a = 1.0 \times 10^{-6}$) would be :

- (1) 10 M
- (2) 10^{-5} M
- (3) 10^{-6} M
- (4) 10^{-10} M

16) **Assertion** : Aquatic species are more comfortable in warm water than in cold water.

Reason : K_H values for both N_2 & O_2 decrease with increase of temperature.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (3) Assertion is True but the Reason is False.
 (4) Both Assertion & Reason are False.

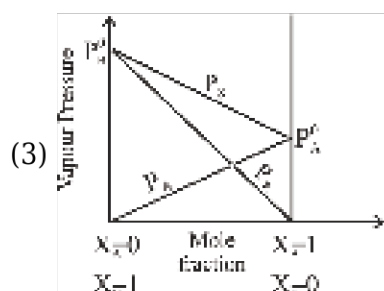
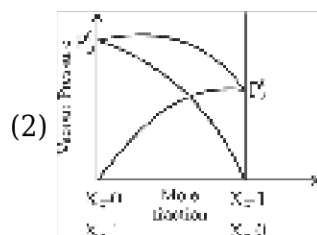
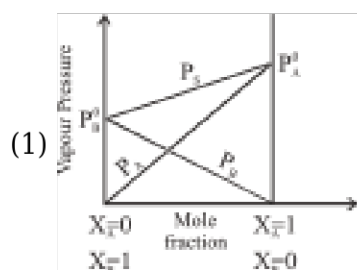
17)

Vapour pressure of solution of benzene

($P^\circ = 120$ torr) and toluene ($P^\circ = 80$ torr) having 2 moles of each is:-

- (1) 400 torr
 (2) 200 torr
 (3) 100 torr
 (4) 50 torr

18) Which of the following graph is correct for an ideal solution given that $P_A^0 < P_B^0$:-



(4) None of these

19) The van't Hoff factor i for an electrolyte which undergoes dissociation and association in solvent are respectively :-

- (1) Greater than one and less than one
 (2) Less than one and greater than one
 (3) Less than one and less than one
 (4) Greater than one and greater than one

20) Consider following statements :-

(a) Azeotropes are mixture having the same composition in liquid and vapour phase.

- (b) It is not possible to separate the components of azeotropes by fractional distillation.
(c) Azeotropic mixture is ideal solution.
(d) Solutions which have large positive deviation from Raoult law form minimum boiling azeotrope.

Which statement is/are correct.

- (1) (a), (b), (c)
(2) (a), (b), (d)
(3) (a), (c)
(4) (c), (d)

21) Which solution have maximum freezing point :

- (1) 1 M CaF_2
(2) 1.5 M $\text{Al}_2(\text{SO}_4)_3$
(3) 2 M NaCl
(4) 1 M AgNO_3

22) As a result of osmosis, the volume of the concentrated solution

- (1) Gradually decreases
(2) Gradually increases
(3) Suddenly increases
(4) Suddenly decreases

23) An example of colligative property is-

- (1) Freezing point
(2) Boiling point
(3) Vapour pressure
(4) Osmotic pressure

24) What will be the degree of dissociation of 0.1 M $\text{Mg}(\text{NO}_3)_2$ solution if van't Hoff factor is 2.74?

- (1) 75%
(2) 87%
(3) 100%
(4) 92%

25) Which of the following statement is correct if the intermolecular force in liquids A, B and C are in the order $A < B < C$?

- (1) B evaporates more readily than A
(2) B evaporates less readily than C
(3) A evaporates more readily than C
(4) A and B evaporate at the same rate

26) Which of the following is correct ?

- (1) Conductivity of a solution decreases with dilution, whereas molar conductivity increases with dilution
- (2) Conductivity of a solution increases with dilution, whereas molar conductivity decreases with dilution
- (3) Both conductivity and molar conductivity decrease with dilution
- (4) Both conductivity and molar conductivity increase with dilution

27) A conductance cell when filled with 0.5 M KCl solution (conductivity = $6.67 \times 10^{-3} \Omega^{-1} \text{cm}^{-1}$) registers a resistance of 243 Ω . Its cell constant is:-

- (1) 1.62 cm
- (2) 1.62 cm^{-1}
- (3) 1.62 dm^{-1}
- (4) 1.62 m^{-1}

28) The $E^\circ_{\text{M}^{3+}/\text{M}^{2+}}$ values for Cr, Mn, Fe and Co are +0.41, +1.57, +0.77 and +0.197 V respectively. For which one of these metals the change in oxidation state from +2 to +3 is easiest :-

- (1) Co
- (2) Mn
- (3) Fe
- (4) Cr

29) Four alkali metals P, Q, R and S are having standard electrode potentials as -3.05, -1.66, -0.40 and 0.80V respectively. Which one will be most reducing ?

- (1) P
- (2) Q
- (3) R
- (4) S

30) When an electric current is passed through acidified water, 112 ml. of hydrogen gas at STP collects at the cathode in 965 seconds. The current passed, in ampere is :

- (1) 1.0
- (2) 0.5
- (3) 0.1
- (4) 2.0

31) The value of Λ^∞ for NH_4Cl , NaOH and NaCl are 150, 250 and $130 \text{ ohm}^{-1} \text{cm}^2 \text{eq}^{-1}$ respectively. The value of Λ^∞ of NH_4OH is :-

- (1) 270
- (2) 180

(3) 400

(4) 380

32) The weight ratio of Al (atomic mass = 27) and Ag (atomic mass = 108) deposited using the same quantity of current is :

(1) 9 : 108

(2) 2 : 12

(3) 108 : 9

(4) 3 : 8

33) The charge required for the reduction of 1 mol of MnO_4^- to MnO_2 , is :

(1) 1F

(2) 3F

(3) 5F

(4) 7F

34) **Statement-I** : The cell potential becomes half if the cell reaction is divided by 2 throughout.

Statement-II : Cell voltage is an extensive quantity.

(1) Both statement I and II are incorrect.

(2) Both statement I and II are correct.

(3) Statement I is incorrect but statement II is correct.

(4) Statement I is correct but statement II is incorrect

35) Emf of the given cell at 298 K is

$\text{Mg(s)} \mid \text{Mg}^{2+} (0.0001 \text{ M}) \parallel \text{Cu}^{+2} (0.001 \text{ M}) / \text{Cu(s)}$

Given $E^\circ \text{Mg}^{2+}/\text{Mg} = -2.36 \text{ V}$

$E^\circ \text{Cu}^{+2}/\text{Cu} = 0.34 \text{ V}$

(1) 2.73 V

(2) 2.67 V

(3) -2.73 V

(4) -2.67 V

SECTION-2

1) For the reaction, $\text{PCl}_{3(g)} \rightleftharpoons \text{PCl}_{3(g)} + \text{Cl}_{2(g)}$

The forward reaction at constant temperature is favoured by :

(1) adding inert gas at constant volume

(2) adding Cl_2 at constant volume

(3) adding inert gas at constant pressure

(4) none

2) For the reaction



If at equilibrium each of PCl_5 , PCl_3 and Cl_2 was found to be 3 mol and the total pressure is 3 atm.

What is the value of K_p ?

- (1) 3.0
- (2) 4.0
- (3) 6.0
- (4) 1.0

3) For which of the following reactions $K_p < K_c$:-

- (1) $\text{N}_2\text{O}_3(\text{g}) \rightleftharpoons \text{NO}_2(\text{g}) + \text{NO}(\text{g})$
- (2) $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{H}_2\text{O}(\text{g})$
- (3) $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$
- (4) $2\text{SO}_3(\text{g}) \rightleftharpoons 2\text{SO}_2(\text{g}) + \text{O}_2(\text{g})$

4) If, $[\text{OH}^-] = 2 \times 10^{-3} \text{ M}$, then pH of solution is :-

- (1) 2.7
- (2) 11.3
- (3) 3.7
- (4) 12.3

5) Conjugate acid of HCO_3^- is :-

- (1) H_2CO_3
- (2) CO_3^{2-}
- (3) CO_2
- (4) None of these

6) The conjugate acid of a strong base is a :-

- (1) Weak acid
- (2) Strong acid
- (3) Strong base
- (4) Weak base

7) Which of the following aqueous solution has pH greater than 7 :-

- (1) CuSO_4
- (2) FeCl_3
- (3) NaCl
- (4) CH_3COONa

8) Mole fraction of A in H_2O is 0.2. The molality of A in H_2O is :

- (1) 13.9
- (2) 15.5
- (3) 14.5
- (4) 16.8

9) In a mixture of A and B, components show positive deviation when :-

- (1) A - B interaction is stronger than A - A and B - B interaction
- (2) A - B interaction is weaker than A - A and B - B interaction
- (3) $\Delta V_{\text{mix}} < 0$, $\Delta S_{\text{mix}} > 0$
- (4) $\Delta V_{\text{mix}} = 0$, $\Delta S_{\text{mix}} > 0$

10) What will be the osmotic pressure of 0.2 M $\text{K}_4[\text{Fe}(\text{CN})_6]$ at 27°C ?

- (1) 24.6 atm
- (2) 49.2 atm
- (3) 4.92 atm
- (4) 2 atm

11) How much volume of 1 M H_2SO_4 is needed to neutralize 10 ml of 1 M NaOH solution :-

- (1) 25 ml
- (2) 5 ml
- (3) 10 ml
- (4) 20 ml

12) Given, $E_{\text{Ag}^+/\text{Ag}}^\circ = 0.80\text{V}$,
 $E_{\text{Mg}^{2+}/\text{Mg}}^\circ = -2.37\text{V}$, $E_{\text{Cu}^{2+}/\text{Cu}}^\circ = 0.34\text{V}$,
 $E_{\text{Hg}^{2+}/\text{Hg}}^\circ = 0.79\text{V}$.

Which of the following statements is correct

- (1) AgNO_3 can be stored in copper vessel.
- (2) $\text{Cu}(\text{NO}_3)_2$ can be stored in magnesium vessel.
- (3) CuCl_2 can be stored in silver vessel.
- (4) HgCl_2 can be stored in copper vessel.

13) Which of the following metals can be displaced by Zn from its aqueous solution :-

- (1) Mg
- (2) Cu
- (3) Na
- (4) Al

14) In a galvanic cell, the electrons flow from :-

- (1) Anode to cathode through the solution
- (2) Cathode to anode through the solution
- (3) Anode to cathode through the external circuit
- (4) Cathode to anode through the external circuit

15) In salt bridge normally KCl is used because :

- (1) It is strong electrolyte
- (2) It is good conductor of electricity
- (3) K^+ and Cl^- ions have nearly same ionic mobility
- (4) It is an ionic compound

BOTANY

SECTION-1

1) In mitochondrial ETS, cytochrome 'C' acts as a mobile carrier for transfer of electrons between:

- (1) Complex III and IV
- (2) Complex I and II
- (3) Complex II and III
- (4) Complex IV and V

2) How many $NADH+H^+$ are formed from one molecule of pyruvic acid in oxidative decarboxylation (link reaction)?

- (1) Two
- (2) Three
- (3) One
- (4) Five

3) Final hydrogen acceptor in aerobic respiration is:-

- (1) O_2
- (2) CO_2
- (3) PGAL
- (4) C_2H_5OH

4) Ubiquinone is located

- (1) On the outer surface of inner membrane
- (2) On the inner surface of outer membrane
- (3) Within inner membrane
- (4) Within outer membrane

5) The first member of the TCA cycle is

- (1) Acetyl CoA
- (2) Citric acid
- (3) OAA
- (4) Pyruvic acid

6) In Krebs' cycle, number of molecules of CO_2 , NADH, FADH_2 and GTP produced from one glucose molecule

- (1) 6 CO_2 , 8 NADH + H^+ , 1 FADH_2 , 1 GTP
- (2) 6 CO_2 , 6 NADH + H^+ , 2 FADH_2 , 2 GTP
- (3) 4 CO_2 , 6 NADH + H^+ , 1 FADH_2 , 1 GTP
- (4) 4 CO_2 , 6 NADH + H^+ , 2 FADH_2 , 2 GTP

7) ETS is present in

- (1) Stroma
- (2) Matrix of mitochondria
- (3) Inner mitochondrial membrane
- (4) Outer membrane of mitochondria

8) How many ATP molecules can be produced through oxidative phosphorylation of 1NADH+ H^+ and 4 FADH_2

- (1) 15
- (2) 24
- (3) 6
- (4) 11

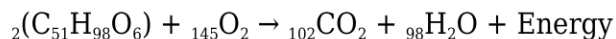
9) Sucrose is converted into glucose and fructose by the enzyme:

- (1) Invertase
- (2) Zymase
- (3) Hexokinase
- (4) Ligase

10) Number of CO_2 molecules evolved in glycolysis is:

- (1) 2
- (2) 1
- (3) 3
- (4) 0

11) What will be the RQ for the following equation.



- (1) 0.9
- (2) 1
- (3) 0.8
- (4) 0.7

12) If fatty acids were to be respired, they would first be degraded to :

- (1) Glucose-6 phosphate
- (2) Pyruvate
- (3) Acetyl CoA
- (4) Alpha ketoglutarate

13) Which of the following is **not** the product released during respiration ?

- (1) CO₂
- (2) Water
- (3) Energy
- (4) O₂

14) The compounds that are subjected to biological oxidation are called _____ in which _____ is the most common.

- (1) Respiratory substrate, carbohydrate
- (2) Respiratory substance, protein
- (3) Respiratory organic substances, fats
- (4) Respiratory substrate, protein

15) When proteins are respiratory substrates the RQ value would be about ?

- (1) 1
- (2) 0.7
- (3) 0.9
- (4) 1.5

16) Pyruvate dehydrogenase activity during aerobic respiration requires :-

- (1) Calcium
- (2) Iron
- (3) Cobalt
- (4) Magnesium

17) Glycolysis scheme was given by which of the following scientist :-

- (1) Gustav Embden

- (2) Otto Meyerhoff
- (3) J. Parnas
- (4) All of these

18) ATP is produced during the light reactions *via*

- (1) Carbon dioxide fixation
- (2) Chemiosmotic hypothesis
- (3) Reduction of water
- (4) All of the above

19) Synthesis of food in C_4 pathway occurs in chloroplast of

- (1) Guard cells
- (2) Bundle sheath
- (3) Spongy mesophyll
- (4) Palisade cells

20) Photolysis of water takes place in

- (1) Calvin cycle
- (2) Glycolysis
- (3) Light phase
- (4) Dark phase

21) $NADP^+$ is converted into $NADPH + H^+$ in:

- (1) Photosystem-I
- (2) Non-cyclic photophosphorylation
- (3) Calvin cycle
- (4) Photosystem-II

22) In cyclic photophosphorylation which one of the following is formed?

- (1) ATP
- (2) NDP and ATP
- (3) $NADH + H^+$ and O_2
- (4) $NADPH + H^+$, ATP and O_2

23) Photosystem-II occurs in

- (1) Stromal lamellae
- (2) Granal lamellae
- (3) One surface of mitochondria
- (4) On the outer membrane of chloroplast

24) At low light intensities, the relation between the incident light and CO_2 fixation rates is

- (1) Sigmoid
- (2) Linear
- (3) Parabola
- (4) Hyperbola

25) Why Calvin cycle is called C_3 -cycle ?

- (1) Primary CO_2 acceptor is 3 carbon - compound.
- (2) Many intermediate compounds are 3 carbon- compounds.
- (3) 1st stable product is 3-PGA which is a 3 carbon- compound.
- (4) None of the above

26) The enzyme RuBisCO is found in

- (1) Chloroplast
- (2) Mitochondria
- (3) Cytoplasm
- (4) Nucleus

27) In PS-I the reaction centre Chl *a* has absorption maxima at _____, while in PS-II the reaction centre Chl *a* has absorption maxima at _____.

- (1) P_{680} , P_{700}
- (2) P_{700} , P_{680}
- (3) P_{800} , P_{600}
- (4) P_{700} , P_{900}

28) Photosystem-I (PS I) and photosystem-II (PS II) are named

- (1) In the sequence of their discovery
- (2) In the sequence in which they function during light reaction
- (3) In the sequence of their positions
- (4) In the sequence of e^- transfer

29) Chemiosmotic hypothesis of ATP synthesis in the chloroplasts is based on the

- (1) Accumulation of K ions
- (2) Proton gradient
- (3) Accumulation of Na ions
- (4) Membrane potential

30) Kranz anatomy is observed in

- (1) C_2 plants

- (2) C_3 plants
- (3) C_4 plants
- (4) CAM plants

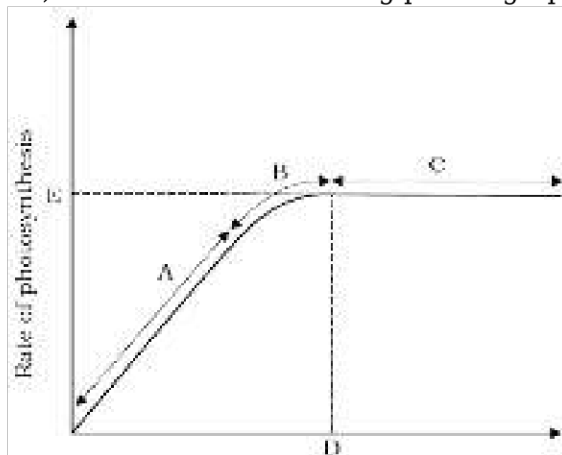
31) How many turns of Calvin cycle yield one molecule of glucose ?

- (1) Eight
- (2) Two
- (3) Six
- (4) Four

32) Which among the following is the major limiting factor for photosynthesis?

- (1) Light
- (2) CO_2
- (3) Temperature
- (4) Water

33) At which of the following part of graph light is limiting factor ?



Light intensity

- (1) D
- (2) E
- (3) C
- (4) A

34) C_4 plants show saturation at about _____ μL^{-1} of CO_2 .

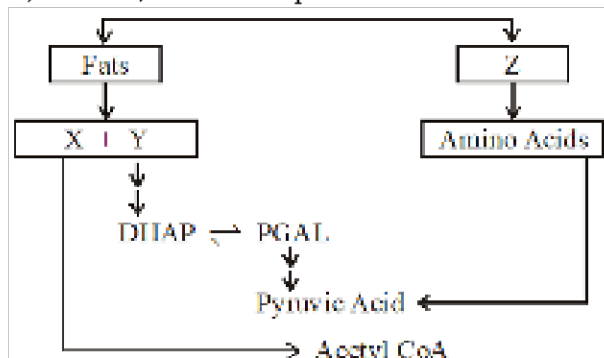
- (1) 450
- (2) 360
- (3) 50
- (4) 10

35) Action spectrum for photosynthesis was first given by Engelmann by using :-

- (1) *Cladophora* and anaerobic bacteria
- (2) *Chlorella* and aerobic bacteria
- (3) *Cladophora* and aerobic bacteria
- (4) BGA and aerobic bacteria

SECTION-2

1) Here X, Y and Z represent :-



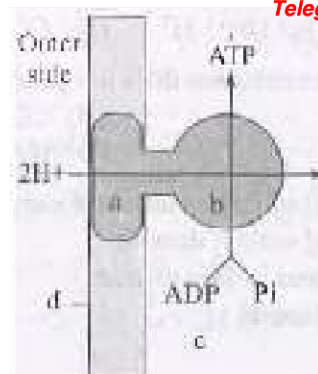
- (1) X-Glycerol, Y-fatty acid, Z-Protein
- (2) X - Fatty acid, Y - Glycerol, Z-Organic acid
- (3) X - Fatty acid, Y - Glycerol, Z-protein
- (4) X - Organic acid, Y - Glycerol, Z-Carbohydrate

2) In Embden, Meyerhof and Parnas pathway respectively at how many steps ATP is synthesised, NAD^+ is reduced and ATP is utilised ?

- (1) Two, One and Two
- (2) One, Two and One
- (3) Five, Two and Two
- (4) Three, One and Two

3) Complex IV refers to cytochrome c oxidase complex containing cytochromes

- (1) b and c_1 and one copper centre
- (2) a and a_3 and four copper centres
- (3) c_1 and c and three copper centres
- (4) a and a_3 and two copper centres



4) Recognise the figure and find out the **correct** matching.

- (1) a— F_1 , b— F_0 , c—intermembrane space, d—outer mitochondrial membrane
- (2) a— F_0 , b— F_1 , c—intermembrane space, d—inner mitochondrial membrane
- (3) a— F_1 , b— F_0 , c—matrix, d—inner mitochondrial membrane
- (4) a— F_0 , b— F_1 , c—matrix, d—inner mitochondrial membrane

5) **Statement-A:** In aerobic respiration, glycolysis is the only process in respiration

Statement-B: Glycolysis occurs in the cytoplasm of the cell and is present in all living organisms.

- (1) Statement-A is correct
- (2) Statement-B is correct
- (3) Both the Statement are correct
- (4) Both the Statement are incorrect

6) **Assertion :** There is very little transport of gases from one plant part to another.

Reason : Each plant part takes care of its own gas exchange needs

- (1) Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false
- (4) Assertion and Reason are false

7) Which among the following is **wrong** ?

- (i) Other than glucose, no other substrates can be used in respiratory process.
- (ii) Respiratory pathway is an amphibolic pathway.
- (iii) In respiratory pathway different substrates would enter if they were to be respired and used to derive energy.

- (1) Only i
- (2) Only iii
- (3) Only ii
- (4) Both i and ii

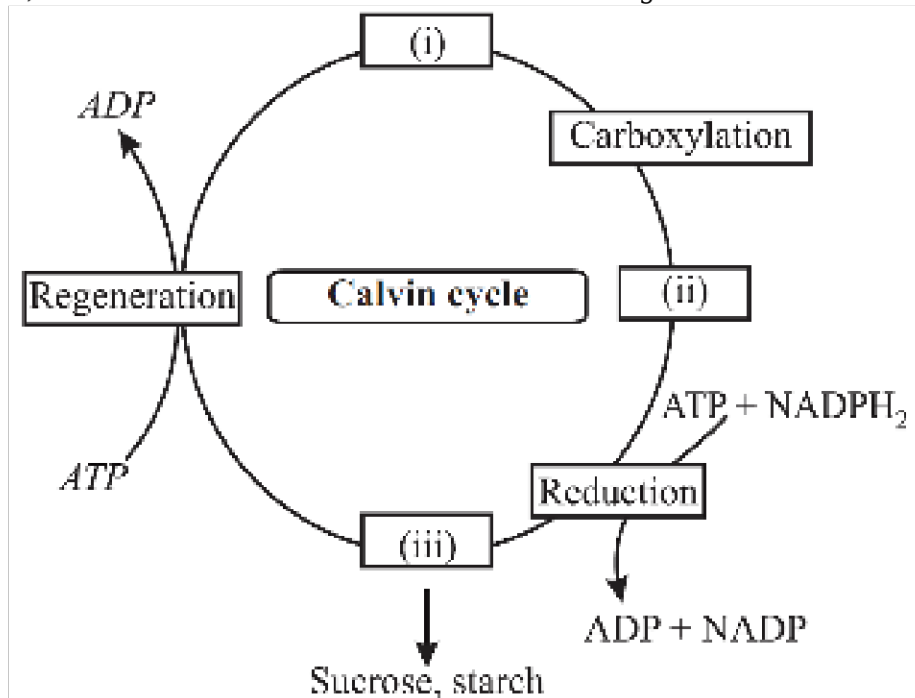
8) The first product of CO_2 fixation in Hatch and Slack pathway in plants is:-

- (1) Formation of malic acid in bundle sheath cells
- (2) Formation of pyruvic acid in mesophyll cells

(3) Formation of OAA in bundle sheath cells

(4) Formation of OAA in the mesophyll cells

9) Choose the **correct** combinations of labelling involved in the Calvin cycle.



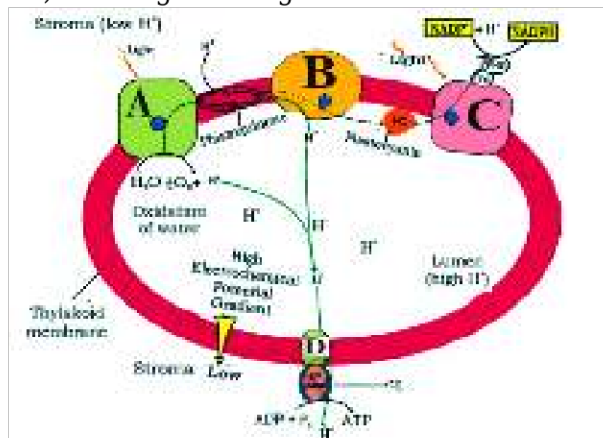
(1) (i) RuBP (ii) PGA (iii) Triose phosphate

(2) (i) PGA (ii) RuBP (iii) Triose phosphate

(3) (i) PGA (ii) Triose phosphate (iii) RuBP

(4) (i) Triose phosphate (ii) PGA (iii) RuBP

10) In this given diagram match the following -



	A	B	C	D
(1)	PS-I	PS-II	Cyt-b ₆ f	F ₁
(2)	PS-II	Cyt-b ₆ f	PS-I	F ₀
(3)	PS-I	Cyt-b ₆ f	PS-I	F ₁
(4)	PS-II	PS-I	Cyt-b ₆ f	F ₀

(1) 1

(2) 2

(3) 3

(4) 4

11) The whole scheme of transfer of electrons, starting from PS-II, Uphill to the acceptor, down the e^- transport chain to PS-I, excitation of electrons, transfer to another acceptor, and finally downhill to $NADP^+$ is called

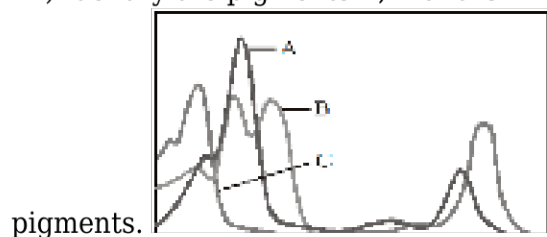
(1) Y-scheme

(2) δ -scheme

(3) Z-scheme

(4) None of these

12) Identify the pigments A, B and C in the given graph showing absorption spectrum of these



pigments.

	A	B	C
(1)	Chlorophyll- <i>a</i>	Carotenoids	Chlorophyll- <i>b</i>
(2)	Chlorophyll- <i>b</i>	Carotenoids	Chlorophyll- <i>a</i>
(3)	Chlorophyll- <i>a</i>	Chlorophyll- <i>b</i>	Carotenoids
(4)	Carotenoids	Chlorophyll- <i>a</i>	Chlorophyll- <i>b</i>

(1) 1

(2) 2

(3) 3

(4) 4

13) If a chemical process is affected by more than one factors, then its rate will be determined by the factor which is nearest to its minimum value, it is the factor which directly affects the process if its quantity is changed. This is the statement of

(1) Hatch and Slack

(2) Calvin and Bassham

(3) Blackman's law of limiting factor

(4) Lindemann's law of minimum

14) **Assertion** : Chlorophyll *a* is the chief pigment associated with photosynthesis.

Reason : Wavelengths at which there is maximum absorption by chlorophyll *b*, i.e., in the blue and the red regions, also shows higher rate of photosynthesis.

(1) Both Assertion & Reason are True and the Reason is the correct explanation of the Assertion

(2) Both Assertion & Reason are True but the Reason is not the correct explanation of the Assertion

- (3) Assertion is True statement but Reason is false
- (4) Both Assertion and Reason are False statements

15) Scientist which performed a series of experiments that revealed the essential role of air in the growth of green plants was

- (1) Joseph Priestley
- (2) Jan Ingenhousz
- (3) Julius von Sachs
- (4) T. W. Engelmann

ZOOLOGY

SECTION-1

1) Consider the following statements.

- (1) *Camelus*, *Delphinus*, *Elephas* and *Struthio* can regulate their body temperature and are placed under same class.
- (2) *Carcharodon*, *Pristis*, *Trygon* and *Panthera* are all oviparous and are placed under same class.
- (3) *Pteropus*, *Macropus*, *Ornithorhynchus* and *Balaenoptera* are viviparous and are placed under same class.
- (4) *Betta*, *Pterophyllum*, *Exocoetus* and *Clarias* have external fertilization and are placed under same class.

2) Read the following statements and choose the correct option.

Statement A: Reptiles have a 3-chambered heart, except crocodiles which have a 2-chambered heart.

Statement B: Superclass Pisces shows single circulation as opposed to class Aves which shows double circulation.

- (1) Both statements are incorrect.
- (2) Only statement A is correct.
- (3) Only statement B is correct.
- (4) Both statements are correct.

3) All living members of class cyclostomata are

- (1) Endoparasites in some fishes
- (2) Ectoparasites on some fishes
- (3) All free living
- (4) Endoparasites in amphibians

4) Torpedo is a member __ I __ and commonly called __ II __ . Choose the correct option to fill in the blanks.

	I	II
(1)	Osteichthyes	Sting ray
(2)	Chondrichthyes	Sting ray
(3)	Chondrichthyes	Electric ray
(4)	Osteichthyes	Electric ray

(1) 1

(2) 2

(3) 3

(4) 4

5) In amphibians, respiration takes place by

(1) Gills

(2) Lungs

(3) Skin

(4) All of these

6) Tree frog is

(1) *Ichthyophis*

(2) *Hyla*

(3) *Rana*

(4) *Bufo*

7) The class name reptile refers to

(1) Presence of air cavities in bone

(2) Presence of internal fertilization

(3) Creeping or crawling mode of locomotion

(4) Refers to presence of tympanum

8) In birds, the digestive tract has additional chambers like

(1) Gizzard and hepatic caecae

(2) Ileum and rectal papillae

(3) Crop and gizzard

(4) Proventriculus and hepatic caecae

9) **Assertion** : Cartilaginous fishes have to swim constantly to avoid sinking.

Reason : These fishes lack air bladder which regulates buoyancy.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

10) Which of the following statement is false about the picture of the following given vertebrate?



- (1) Possess circular and sucking mouth.
- (2) They are marine but migrate for spawning to fresh water.
- (3) Possess 4 pair of gill slits
- (4) Devoid of scales.

11) Limbless reptiles :-

- (1) *Ichthyophis*
- (2) Snakes
- (3) *Salamandra*
- (4) Both (1) and (2)



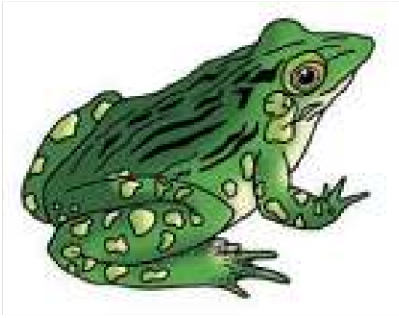
12) Which of the following is a correct feature for animal?

- (1) It has four pairs of gills without operculum
- (2) It has placoid scale
- (3) Heart is three chambered
- (4) It has air bladder

13) Identify the animal that belong to the described class :-

- (a) Having terminal mouth
- (b) Four pairs of gills
- (c) Gills covered by operculum

- (1) *Pristis*
- (2) *Exocoetus*
- (3) *Hyla*
- (4) *Petromyzon*



14)

Identify the statements which are true to the given animal :-

- (a) Body divisible into head and trunk.
- (b) Moist skin without scales.
- (c) Eyes do not have eyelids.
- (d) Cloaca is absent.

- (1) (a) and (d)
- (2) (b) and (c)
- (3) (c) and (d)
- (4) (a) and (b)

15) Complete the analogy and choose the correct option.

Poikilothermous : *Chelone* :: Homeothermous : _____.

- (1) *Aptenodytes*
- (2) *Alligator*
- (3) *Bufo*
- (4) *Clarias*

16) How many digits are found in forelimbs and hindlimbs of frog respectively ?

- (1) 5, 5
- (2) 5, 4
- (3) 4, 4
- (4) 4, 5

17) Vocal sac & Copulatory pad are the characteristics of :-

- (1) Male frog
- (2) Female frog
- (3) Both (1) & (2)
- (4) Male cockroach

18) Bidder's canal in frog is found in ?

- (1) Liver
- (2) Kidney
- (3) Testis
- (4) Ovary

19) **Statement I :** In frog each kidney is composed of several structural and functional units called uriniferous tubules

Statement II : In frog forebrain includes olfactory lobes, paired cerebral hemispheres and paired diencephalon.

- (1) Both statements I and II are correct
- (2) Both statements I and II are incorrect
- (3) Only statement II is correct
- (4) Only statement I is correct

20) How many ova are laid by mature female frog at a time ?

- (1) 2000-5000
- (2) 2500-5000
- (3) 2500-3000
- (4) 100-500

21) The length of alimentary canal in adult frog is :-

- (1) Longer as it is a carnivore.
- (2) Longer as it is a herbivore.
- (3) Shorter as it is a carnivore.
- (4) Shorter as it is a herbivore.

22) In frog, RBCs are:

- (1) Nucleated and biconcave
- (2) Enucleated and biconvex
- (3) Nucleated and biconvex
- (4) Non-nucleated and biconcave

23) The skin of frog is smooth & slippery due to the presence of ?

- (1) Sebaceous gland
- (2) Mucus gland
- (3) Poisonous gland
- (4) Mammary gland

24) How many vasa efferentia arise from testes in frog:-

- (1) 10 - 12
- (2) 10 - 14
- (3) 9 - 10
- (4) 8 - 10

25) Aestivation can be seen in frog during :-

- (1) Winter
- (2) Summer
- (3) Rainy season
- (4) Autumn

26) The secretion of which gland helps in the lubrication of penis?

- (1) Seminal vesicle
- (2) Prostate
- (3) Bulbourethral
- (4) Epididymis

27) How many structures in the list given below have $(22 + x)$ chromosome ?
Polar Body, Oogonia, Primary Oocyte, Ovum, Secondary oocyte :-

- (1) Four
- (2) Three
- (3) Two
- (4) One

28) Find the **incorrect** statement?

- (1) Head of sperm contains diploid nucleus
- (2) Middle piece have mitochondria
- (3) Sperm motility is essential for fertilisation .
- (4) Both (2) & (3)

29) At what stage does the primary oocyte complete its first meiotic division?

- (1) Primary follicle stage
- (2) Secondary follicle stage
- (3) Tertiary follicle stage
- (4) Antrum stage

30) Spermiogenesis involves :-

- (1) Formation of secondary spermatocyte from primary spermatocyte.
- (2) Formation of spermatids
- (3) Formation of sperms from spermatids
- (4) Formation of primary spermatocytes.

31) The testis are situated outside the abdominal cavity within a pouch called scrotum, because :

- (1) In scrotum, $2 - 2.5^{\circ}\text{C}$ lower than the normal internal body temperature.
- (2) In scrotum, $2 - 2.5^{\circ}\text{C}$ higher than the normal internal body temperature
- (3) In scrotum, $2 - 2.5^{\circ}\text{F}$ higher than the normal internal body temperature

(4) In scrotum, 2 - 2.5°F less than the normal internal body temperature

32) **Assertion** : Seminiferous tubule is a structural and functional unit of testis.

Reason : Each testis has about 250 compartments called testicular lobules and each lobule contains one to three highly coiled seminiferous tubules and in which sperms are produced.

- (1) Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) If both Assertion & Reason are False.

33) Ejaculatory duct is formed by uniting with a duct from which gland?

- (1) Seminal vesicle
- (2) Prostate
- (3) Vasa efferentia and seminal vesicle
- (4) Bulbourethral

34) The _____ lead to vas deferens that ascends to the _____ and loops over the _____ .

- (1) prostate gland, stomach, urinary bladder.
- (2) epididymis, abdomen, urinary bladder.
- (3) vasa efferentia, abdomen, ureter.
- (4) urinary bladder, ejaculatory duct, abdomen.

35) Increased secretion of which hormone starts the process of sperm formation at the time of puberty ?

- (1) GH
- (2) TSH
- (3) PRL
- (4) GnRH

SECTION-2

1) The womb opens into the vagina through _____.

- (1) cervix
- (2) hymen
- (3) clitoris
- (4) ampulla

2) Which of the following is not a part of the female external genitalia?

- (1) Labia majora
- (2) Labia minora

(3) Uterus

(4) Clitoris

3) The part of fallopian tube near uterus is called _____ and near ovary is called _____

(1) Uterus ; Isthmus

(2) Cervical canal ; Isthmus

(3) Isthmus ; Ampulla

(4) Isthmus; Infundibulum

4) **Assertion** : Only one sperm can fertilize an ovum

Reason : During fertilization, a sperm comes in contact with the zona pellucida layer of ovum and induces changes in the membrane that blocks the entry of additional sperm.

(1) Both assertion and reason are true & reason is the correct explanation of assertion.

(2) Both assertion and reason are true & reason is not the correct explanation of assertion.

(3) Assertion is correct but reason is incorrect

(4) Both assertion & reason are incorrect

5) The layer of ovum from inside to outside is _____

(1) Corona radiata → Vitelline membrane → Zona pellucida

(2) Vitelline membrane → Zona pellucida → Corona radiata

(3) Corona radiata → Zona pellucida → Vitelline membrane

(4) Zona pellucida → Corona radiata → Vitelline membrane

6) **Statement I**: The primary follicles get surrounded by more layers of granulosa cells, and new theca and shows fluid filled cavity antrum. Now it is called secondary follicle.

Statement II: Graffian follicle ruptures to release the secondary oocyte from the ovary by the process called ovulation.

(1) Statement I is correct but Statement II is false.

(2) Statement I is incorrect but Statement II is true

(3) Both Statement I and Statement II are true

(4) Both Statement I and Statement II are false

7) **Assertion (A)** :- Secretions of epididymis,

vas deferens, seminal vesicles and prostate are essential for maturation and motility of sperms.

Reason (R) :- The functions of male sex accessory ducts and glands are maintained by the testicular hormones androgens.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion

(3) Assertion is correct but Reason is not correct.

(4) Assertion is not correct but Reason is correct.

8) Read the following statements.

Statement-A: A normal human female has approximately 60,000 - 80,000 primary follicles in both the ovaries at puberty.

Statement-B: A couple of million oogonia are formed in the ovary after birth.

Choose the correct option w.r.t. the above given statements.

- (1) Both the statements are correct
- (2) Both the statements are incorrect
- (3) Only statement A is correct
- (4) Only statement B is correct

9) One primary spermatocyte : 4 sperms :: one primary oocyte : ?

- (1) 4 ova
- (2) 3 ova
- (3) 2 ova
- (4) 1 ovum

10) The sperms need to swim to the ovum for normal fertilisation. The structure which provides the energy source for swimming is present in the

- (1) Acrosome
- (2) Head
- (3) Middle piece
- (4) Tail

11) In human females, meiosis-II is not completed until ?

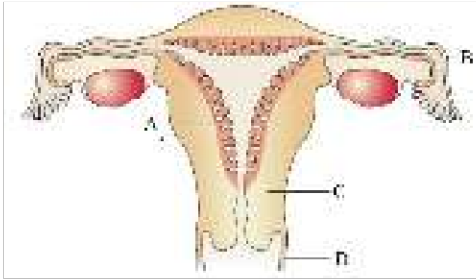
- (1) Birth
- (2) Puberty
- (3) Fertilization
- (4) Uterine implantation

12) **Statement A :** All copulations do not lead to fertilization and pregnancy.

Statement B : Fertilization is possible only if ovum and sperms are transported simultaneously to the ampullary isthmic junction.

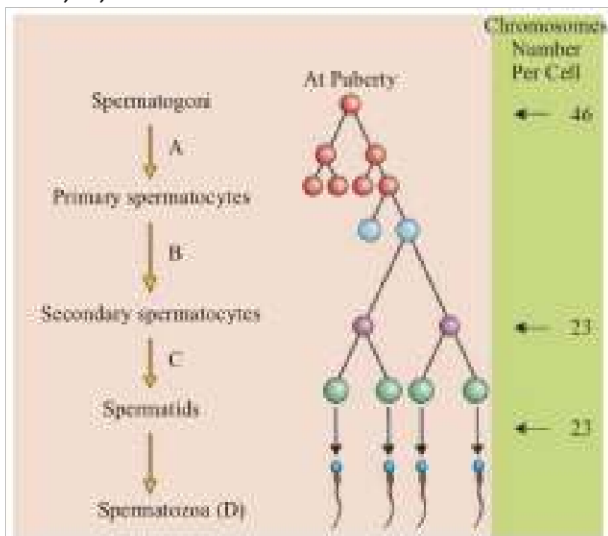
- (1) A is correct and B is incorrect statement
- (2) Both A and B statements are correct
- (3) Only B statement is correct
- (4) Both A and B are incorrect

13) The given diagram shows a human female's reproductive system. Select the part (labelled as A, B, C & D) where semen is released into the female by the penis during copulation.



- (1) A
- (2) B
- (3) C
- (4) D

14) The figure given below shows a flowchart on spermatogenesis. Identify the correct label marked as A, B, C and D.



- (1) A : 1st meiotic division; B : 2nd meiotic division; C : Differentiation; D : 23.
- (2) A : 2nd meiotic division; B : Differentiation; C : 1st meiotic division; D : 46.
- (3) A : Differentiation; B : 2nd meiotic division; C : 1st meiotic division; D : 46.
- (4) A : Mitosis differentiation; B : 1st meiotic division; C : 2nd meiotic division; D : 23.

15) Seminal plasma, the fluid part of semen, is contributed by :

- i. Seminal vesicles
- ii. Prostate
- iii. Urethra
- iv. Bulbourethral glands

- (1) i and ii
- (2) i, ii and iv
- (3) ii, iii and iv
- (4) i and iv

PHYSICS

SECTION-1

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	1	4	2	1	2	4	4	2	2	1	1	4	1	3	2	1	2	2	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35					
A.	4	1	1	4	1	1	4	4	3	3	4	3	1	3	3					

SECTION-2

Q.	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
A.	2	3	2	4	2	1	1	3	1	2	4	4	3	2	1

CHEMISTRY

SECTION-1

Q.	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70
A.	1	3	3	1	3	1	1	2	3	1	3	3	2	2	3	4	3	3	1	2
Q.	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85					
A.	4	2	4	2	3	1	2	1	1	1	1	1	2	1	1					

SECTION-2

Q.	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
A.	3	4	2	2	1	1	4	1	2	1	2	3	2	3	3

BOTANY

SECTION-1

Q.	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	1	3	1	3	3	4	3	4	1	4	4	3	4	1	3	4	4	2	2	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135					
A.	2	1	2	2	3	1	2	1	2	3	3	2	4	2	3					

SECTION-2

Q.	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	1	4	4	2	1	1	4	1	2	3	2	3	2	1

ZOOLOGY

SECTION-1

Telegram: @NEETxNOAH

Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	3	2	3	4	2	3	3	1	3	2	4	2	4	1	4	1	2	4	3
Q.	171	172	173	174	175	176	177	178	179	180	181	182	183	184	185					
A.	3	3	2	1	2	3	2	1	3	3	1	1	1	2	4					

SECTION-2

Q.	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200
A.	1	3	4	1	2	2	2	2	4	3	3	2	4	4	2

PHYSICS

4) Due to conducting sphere

At centre, electric field $E = 0$

And electric potential $V = \frac{Q}{4\pi\epsilon_0 R}$

11) Potential gradient $x = \frac{V}{\ell} = \frac{IR}{\ell} = \frac{I\rho}{A}$

15) $E = 0$ for both +ve & -ve outside sphere & $v=0$

17) $p_1 = q\ell = p_2$ and $P_R = \sqrt{3}q\ell$

27) Initial charges : $q_1 = \sigma(4\pi R^2)$, $q_2 = \sigma(4\pi(2R)^2)$

Let final charges be Q_1 and Q_2 , then,

$$Q = Q_1 + Q_2 = q_1 + q_2 = 5\sigma(4\pi R^2) \dots (i)$$

In final state,

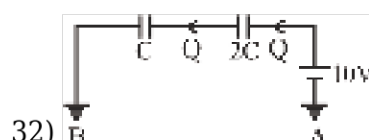
$$V_1 = V_2 \Rightarrow \frac{KQ_1}{R} = \frac{KQ_2}{2R} \Rightarrow \frac{Q_1}{Q_2} = \frac{1}{2} \dots (ii)$$

From (i) and (ii)

$$Q_1 = \frac{Q}{3} \text{ and } Q_2 = \frac{2Q}{3}$$

$$\Rightarrow \sigma_1 = \frac{5\sigma(4\pi R^2)}{3(4\pi R^2)} ; \sigma_2 = \frac{2(5\sigma(4\pi R^2))}{4\pi(2R)^2}$$

$$\Rightarrow \sigma_1 = \frac{5\sigma}{3} \quad \sigma_2 = \frac{5\sigma}{6}$$



$$V_A + 10 - \frac{Q}{2C} - \frac{Q}{C} = V_B$$

$$0 + 10 = \frac{Q}{C} \left[1 + \frac{1}{2} \right] + 0$$

$$10 = \frac{Q}{6\mu} \times \frac{3}{2}$$

$$Q = 40 \mu C$$

33)

$$34) H = \frac{1}{2} \frac{C_1 C_2}{C_1 + C_2} \times (V_1 - V_2)^2$$

$$= \frac{C}{4} (V_1 - V_2)^2$$

$$38) \frac{R_1 R_2}{R_1 + R_2} = \frac{8}{3} \text{ \& } R_1 + R_2 = 12 \Rightarrow R_1 R_2 = 32$$

$$\Rightarrow R_2 - R_1 = \sqrt{(R_1 + R_2)^2 - 4R_1 R_2}$$

$$= \sqrt{12^2 - 4 \times 32} = 4\Omega$$

So, $R_1 = 4\Omega$ and $R_2 = 8\Omega$

$$\text{Hence } \frac{\ell_1}{\ell_2} = \frac{4}{8} = \frac{1}{2}$$

40) Heat given = heat taken

v^2

$$\frac{R}{t} = msdT = \text{const}$$

$v = \text{const.}$ then $t \propto R$

$$\text{Hence in parallel combination } R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

$$\text{then } t = \frac{t_1 t_2}{t_1 + t_2} = \frac{4(6)}{4+6} = 2.4 \text{ min } \textbf{OR } P \times t = Q \text{ (total heat required) which same in all cases.}$$

$$\square P_1 = \frac{Q}{t_1} \text{ and } P_2 = \frac{Q}{t_2}$$

When both are used simultaneously then

$$t = \frac{Q}{P_1 + P_2} = \frac{Q}{\frac{Q}{t_1} + \frac{Q}{t_2}} = \frac{t_1 t_2}{t_1 + t_2} = \frac{4 \times 6}{4 + 6} = 2.4 \text{ min.}$$

42)

Slope of the graph = Resistance

$$R_1 > R_2$$

$$\therefore T_1 > T_2$$

43) As the resistance is divided into 20 pieces, so resistance of each piece = $R/20$

Half of them are joined in series.

□ Effective resistance,

$$R_1 = \frac{R}{20} + \frac{R}{20} + \dots + 10 \text{ times}$$

$$= 10 \times \frac{R}{20} = \frac{R}{2} \dots (1)$$

Remaining half are joined in parallel

$$\square \quad \frac{1}{R_2} = \frac{1}{R/20} + \frac{1}{R/20} + \dots + 10 \text{ times}$$

$$\text{or} \quad \frac{1}{R_2} = \frac{20}{R} + \frac{20}{R} + \dots 10 \text{ times}$$

$$\text{or} \quad \frac{1}{R_2} = \frac{20}{R} \times 10 = \frac{200}{R}$$

$$\square \quad R_2 = \frac{200}{R}$$

As R_1 and R_2 are joined in series, hence,

$$R_{\text{eff.}} = R_1 + R_2 = \frac{R}{2} + \frac{R}{200} = \frac{101R}{200}$$

47)

Relation between electric field and potential difference is given by

$$V_B - V_A = - \int_{r_A}^{r_B} \vec{E} \cdot d\vec{R}$$

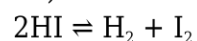
$$V_B - V_A = - \int_{(10,3,0)}^{(4,0,0)} (5000\hat{i}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k})$$

$$V_B - V_A = - \int_{(4,0,0)}^{(10,3,0)} 5000dx = -5000[x]_4^{10}$$

$$= -5000 [10 - 4] = -30 \text{ kV}$$

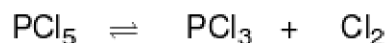
CHEMISTRY

53) Endothermic



(g) (g) (g)

54) NCERT Pg. # 208, Part-1



2 0 0

$$57) \frac{2 \times 60}{100} \quad \frac{2 \times 40}{100} \quad \frac{2 \times 40}{100}$$

$\square$ volume of container = 2 litre

$$K_c = \frac{\left(\frac{2 \times 40}{100 \times 2} \right)^2}{\frac{2 \times 60}{100 \times 2}}$$

$$\therefore K_c = 0.267$$

61) Salt of strong acid and weak base $[(\text{NH}_4)_2\text{SO}_4]$ on hydrolysis form acidic solution.

67)

$$P_m = X_B P_B^0 + X_T P_T^0 = \frac{1}{2} \times 120 + \frac{1}{2} \times 80 = 100 \text{ torr}$$

$$68) P_S = (P_A^0 - P_B^0) \times X_A + P_B^0$$

$$y = mx + C$$

$$\text{given } P_A^0 < P_B^0$$

So slope is negative

70)

Azeotropes mixtures are those which have same composition in liquid and vapour phase they can not be separated by fractional distillations. non ideal solution with positive deviation from Raoult law form minimum boiling azeotrope.

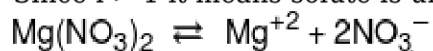
74)

Given

Molarity = 0.1 M

Van't Hoff factor (i) = 2.74

Since i > 1 it means solute is undergoing dissociation



$$\text{Degree of dissociation } (\alpha) = \frac{i-1}{n-1} = \frac{2.74-1}{3-1} = 0.87$$

It means 87%

81)

$$\begin{aligned} \Lambda^\infty \text{NH}_4\text{OH} &= \Lambda^\infty \text{NH}_4\text{Cl} + \Lambda^\infty \text{NaOH} - \Lambda^\infty \text{NaCl} \\ &= 150 + 250 - 130 \\ &= 270 \text{ ohm}^{-1} \text{cm}^{-1} \text{eq}^{-1} \end{aligned}$$

84)

Statement-I : EMF of cell is intensive property

Statement-II : EMF of cell = cell voltage which is intensive

$$93) m = \frac{0.2}{1-0.2} \times \frac{1000}{18} = 13.9$$

95)

$$\pi = iCRT = 5 \times 0.2 \times 0.0821 \times 300$$

97)

Reduction potential of copper is less than silver so tendency to gets reduced is less and hence CuCl_2 can be stored in silver vessel.

BOTANY

113) NCERT XI Pg # 155

115) NCERT Pg # 164

135) NCERT Pg. # 137

ZOOLOGY

160) NCERT - XI, Pg no.- 47

161) NCERT Pg. # 53

163)

NCERT Pg. # 48

164) NCERT Pg.# 48

168) NCERT Pg. # 119

170) NCERT Pg. # 120

173) NCERT Pg No. # 80

174) NCERT Pg. # 119

175) NCERT-XI Bio(E) Pg. # 116-Para No.-2, Line-10

177) NCERT (XII) Pg. # 49 (E) (old NCERT)

178)

[NCERT pg. 48] (old NCERT)

179)

[NCERT pg. 48] (old NCERT)

191)

NEW NCERT pg. # 32 - 33

192) NCERT Pg. # 28

193) NCERT Reference: Biology-XII, Page No.- 48

194) NCERT Page No.- 33

195) NCERT Reference: Biology-XII, Page No.- 48

196) Allen Module Page No.- 55

197) NCERT XII, Page No.- 35

198) Ncert Pg No: 35

199) Ncert Pg No: 33

200) NCERT pg no. 28